Lecture (10) : Oxygen

Oxygen in the form of dioxygen, O_2 (The name “oxygen” means “acid former) that comprises about 21% of the earth's atmosphere Although oxygen is widespread in the Earth's atmosphere and, combined with other elements, in the Earth's crust (which contains 46% oxygen by mass) most of the oxygen on earth is present as oxide and oxoanion minerals (in silicates, carbonates, sulfates, and so forth). Indeed, oxygen combines with almost every element—only some of the noble gases have no known oxygen compounds. the pure element was not isolated and characterized until the 1770s by Karl Wilhelm Scheele and J. Priestley.

Liquid O_2 is used as an oxidizer for rocket fuels. O_2 also is used in the health fields for oxygen-enriched air.

Properties of Oxygen:

The common form of the element oxygen is dioxygen, O_2. (The element also exists as the allotrope ozone, O_3.)

oxygen, is a colorless, odorless gas under standard conditions. The critical temperature is 118 °C. Therefore, you can liquefy oxygen if you first cool the gas below this temperature and then compress it.

Both liquid and solid O_2 have a pale blue color.

The melting point of the solid is 218 °C and the boiling point at 1 atm is 183°C.

Oxygen is only slightly soluble in water; only about 0.04 gram dissolves in 1 liter of water at 25°C.

Preparation of Oxygen:

Oxygen can be prepared in small quantities by decomposing certain oxygen containing compounds. Both the Swedish chemist Karl Wilhelm Scheele and the British chemist Joseph Priestley are credited with the discovery of oxygen. Priestley obtained the gas in 1774 by heating mercury(II) oxide with sunlight.

$$2\,HgO_{(s)} \longrightarrow 2\,Hg_{(l)} + O_{2(g)}$$

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In one laboratory preparation, potassium chlorate, $KClO_3$, is heated with pure manganese(IV) oxide, MnO_2, as a catalyst.

$$2KClO_{3(s)} = 2KCl_{(s)} + 3O_{2(g)}$$

Reactions of Oxygen :

Molecular oxygen is a very reactive gas and combines directly with many substances. Oxygen reacts with almost all elements to give oxides or, in some cases, peroxides or super oxides.

An oxide is a binary compound with oxygen in the $^{-}2$ oxidation state.

Most metals react readily with oxygen to form oxides, especially if the metal is in a form that exposes sufficient surface area.

For example, magnesium wire and iron wool burn brightly in air to yield the oxides.

$$\underset{(s)}{2\,Mg} + \underset{(g)}{O_2} \longrightarrow \underset{(s)}{2\,Mgo}$$

$$\underset{(s)}{3\,Fe} + \underset{(g)}{2\,O_2} \longrightarrow \underset{(s)}{Fe_3O_4}$$

The resulting oxides MgO and Fe_3O_4 are basic oxides, as is true of most metal oxides. If the metal is in a high oxidation state, however, the oxide may be acidic. For example, chromium(III) oxide, Cr_2O_3 is a basic oxide, but chromium(VI) oxide, CrO_3, is an acidic oxide.

The alkali metals form an interesting series of binary compounds with oxygen.

When an alkali metal burns in air, the principal product with oxygen depends on the metal. With lithium, the product is the basic oxide, Li_2O. With the other alkali metals, the product is predominantly the peroxide and superoxide.

A peroxide is a compound with oxygen in the -1 oxidation state. (Peroxides contain either the O_2^{2-} ion or the covalently bonded group (– O – O -).

Sodium metal burns in air to give mainly the peroxide.

$$2Na_{(s)} + O_{2(g)} = Na_2O_{2(s)}$$

A superoxide is a binary compound with oxygen in the -1/2 oxidation state; superoxides contain the superoxide ion, O^{2-}.

Potassium and the other alkali metals form mainly the superoxides.

$$K_{(s)} + O_{2(g)} = KO_{2(s)}$$

Nonmetals react with oxygen to form covalent oxides, most of which are acidic.

For example, carbon burns in an excess of oxygen to give carbon dioxide, which is the acid anhydride of carbonic acid (that is carbon dioxide produces carbonic acid when it reacts with water).

$$C_{(s)} + O_{2(g)} = CO_{2(g)}$$

Sulfur, S8 burns in oxygen to give sulfur dioxide, SO2

the acid anhydride of sulfurous acid.

$$S_{8(s)} + 8O_{2(g)} = 8SO_{2(g)}$$

Sulfur forms another oxide, sulfur trioxide, SO3 but only small amounts are obtained during the burning of sulfur in air. Sulfur trioxide is the acid anhydride of sulfuric acid.

Compounds in which at least one element is in a reduced state are oxidized by oxygen, giving compounds that would be expected to form when the individual elements are burned in oxygen. For example, a hydrocarbon such as octane, C8H18, burns to give carbon dioxide and water.

$$2C_8H_{18(l)} + 25O_{2(g)} = 16CO_{2(g)} + 18H_2O_{(g)}$$

Some other examples are given in the following equations:

$$2H_2S_{(g)} + 3O_{2(g)} = 2H_2O_{(g)} + 2SO_{2(g)}$$

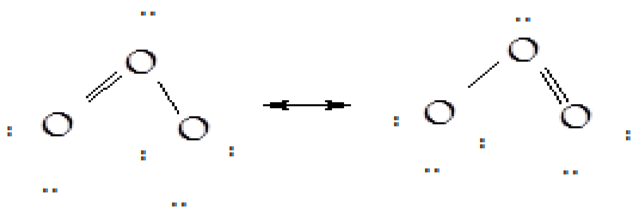
$$CS_{2(l)} + 3O_{2\ (g)} = CO_{2(g)} + 2SO_{2(g)}$$

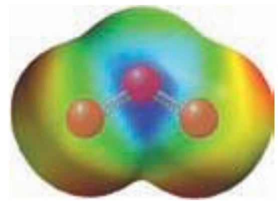
$$2ZnS_{(s)} + 3O_{2(g)} = 2ZnO_{(s)} + 2SO_{2(g)}$$

Note the products that are formed; sulfur compounds usually form SO_2.

Ozone (also known as trioxygen, O_3) :

ozone gas has a characteristic pungent odor and an unstable pale blue gas at room temperature. its density is about 1.5 times that of O_2. At _112°C it condenses to a deep blue liquid. is a reactive form of oxygen having a bent molecular geometry, as you





would predict from its resonance formulas.

On bright sunny days, you might notice the fresh-air smell of dilute ozone.

Although ozone occurs at ground level and is an ingredient of photochemical smog, it is an essential component of the stratosphere, which occurs at about 10 to 15 km above ground level. Ozone in the stratosphere absorbs ultraviolet radiation between 200 and 300 nm, which is vitally important to life on earth. Radiation from the sun contains ultraviolet rays that are harmful to the DNA of biological organisms. Ozone, O_3, is such a powerful oxidizing agent that in significant concentrations it destroys many plastics, metals, and rubber, as well as both plant and animal tissues.

ozone plays a very important role in the absorption of harmful radiation from the sun.The high-altitude ozone layer is responsible for absorbing much of the dangerous ultraviolet light from the sun in the 20–30 Å wavelength range.

Preparation of Ozone:

Ozone forms in the atmosphere whenever O_2 is irradiated by ultraviolet light or subjected to electrical discharges.

$$3O_{2(g)} = 2O_{3(g)}$$

Ozone is formed in the upper atmosphere as some O_2 molecules absorb high-energy electromagnetic radiation

from the sun and dissociate into oxygen atoms; these then combine with other O_2 molecules to form ozone.

$$O_{2(g)} + \text{UV radiation} = 2O_{(g)} \quad \text{(step 1—occurs once)}$$

$$O2(g) + O_{(g)} = O_{3(g)} \quad \text{(step 2—occurs twice)}$$

$$3O_{2(g)} + \text{UV radiation} = 2O_{3(g)} \quad \text{(net reaction)}$$

Reaction of Ozone:

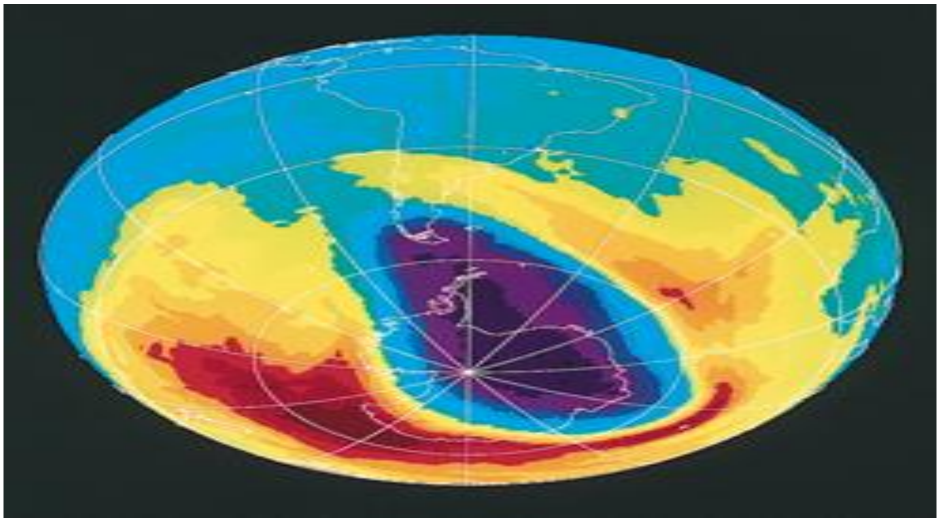
reaction that occurs in the decomposition of ozone in the stratosphere by nitrogen monoxide.

$$O_3 + NO = O_2 + NO_2$$

Spontaneously decomposes.

$$O_3{}^* = O_2 + O$$

$$2O_{3(g)} = 3O_{2(g)}$$



A computer-generated image of part of the Southern Hemisphere on October 17, 1994, reveals the ozone “hole” (black and purple areas) over Antarctica and the tip of South America. Relatively low ozone levels (blue and green areas) extend into much of South America as well as Central America. Normal ozone levels are shown in yellow, orange, and red. The ozone hole is not stationary but moves about as a result of air currents. (Courtesy NASA)

By September 1992, this ozone hole was nearly three times the area of the United States. In December 1994, three years of data from NASA’s Upper Atmosphere Research Satellite (UARS) provided conclusive evidence that CFCs are primarily responsible for this destruction of the ozone layer